**Jackfruit Phase-1**

# Software Requirements Specification

for

# Online Bookstore

**Version 1.0**

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**TABLE OF CONTENTS**


1. Introduction ..... 3
    1.1 Purpose ..... 3
    1.2 Scope ..... 3
    1.3 Audience ..... 4
    1.4 Definitions ..... 5
    1.4 References ..... 6
2. Overall Description ..... 6
    2.1 Product Perspective ..... 6
    2.2 Major Product Functions (Detailed) ..... 7
    2.3 User Roles and Characteristics (Expanded) ..... 7
    2.4 Operating Environment ..... 7
    2.5 Constraints ..... 8
3. External Interface Requirements ..... 8
    3.1 User Interfaces ..... 8
    3.2 Hardware Interfaces ..... 9
    3.3 Software Interfaces ..... 9
    3.4 Communications Interfaces ..... 9
    Security Objectives ..... 10
    Security Requirements ..... 10
4. Analysis Models ..... 11
    4.1 Use Case Model ..... 11
    4.2 Domain Model ..... 11
    4.3 Activity Model ..... 12
    4.4 Sequence Model ..... 12
5. System Features ..... 12
    5.1 Customer Account Management ..... 13
    5.2 Book Catalogue and Search ..... 13
    5.3 Shopping Cart ..... 13
    5.4 Checkout and Delivery ..... 13
    5.5 Payment Processing ..... 13
    5.6 Order Management ..... 13
    5.7 Inventory Management ..... 14
    5.8 Administrator Management ..... 14

**5.9 Order Tracking and Status ..... 14**
**5.10 Security and Access Control ..... 14**
**Security Objectives ..... 14**
**Security Requirements ..... 14**
**6. Non-Functional Requirements ..... 15**
**6.1 Performance Requirements ..... 15**
**6.2 Scalability Requirements ..... 15**
**6.3 Reliability and Availability Requirements ..... 15**
**6.4 Usability Requirements ..... 16**
**6.5 Maintainability and Supportability Requirements ..... 16**
**6.6 Portability Requirements ..... 17**
**7. Other Requirements ..... 17**
**7.1 Legal and Regulatory Requirements ..... 17**
**7.2 Database Requirements ..... 17**
**7.3 Business Rules ..... 18**
**7.4 Assumptions and Dependencies ..... 18**

**Revision History**

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## 1. Introduction

### 1.1 Purpose

**This Software Requirements Specification (SRS) defines the functional and non-functional requirements for Version 1.0 of the Online Bookstore. The**

**system is a web-based platform that enables customers to search, browse, purchase, and track books online.**

**This document specifies the expected behaviour of the system, including customer account management, book catalogue and search, shopping cart, checkout, payment processing, order management, order tracking, inventory management, and administrative operations.**

**The SRS serves as a reference for the development team, testers, stakeholders, and evaluators during system design, implementation, testing, and validation.**

### 1.2 Scope

**The Online Bookstore is a web-based e-commerce platform that provides customers with a centralized system for discovering and purchasing books.**

**The system allows customers to:**

- **Register and manage their accounts.**
- **Log in and authenticate securely.**
- **Browse and search the book catalogue.**
- **Filter and sort books based on relevant attributes.**
- **View detailed information about books.**
- **Add, update, and remove items from the shopping cart.**
- **Checkout and provide delivery information.**
- **Make payments using available payment methods.**
- **Receive order confirmation and an order ID.**
- **View previous and current orders.**
- **Track order status.**
- **Cancel orders when permitted by the applicable business rules.**

**The system also provides administrators with facilities to:**

- **Add, update, and remove books.**
- **Manage book information and pricing.**
- **Manage inventory and stock quantities.**
- **View and manage customer orders.**
- **Update order statuses.**
- **Prevent purchases when sufficient stock is unavailable.**

**In Scope: Customer management, catalogue management, search, cart, checkout, payment, order management, tracking, inventory management, administration, security, and system quality requirements.**

**Out of Scope: Physical warehouse operations, book publishing, physical delivery operations, banking infrastructure, and internal implementation of third-party payment or courier services.**

### 1.3 Audience

**This document is intended for:**

- **Development Team: To understand system functionality, constraints, and implementation requirements.**
- **System Designers: To design the application architecture, database, interfaces, and system modules.**
- **Testing/QA Team: To derive test cases and verify functional and non-functional requirements.**
- **Administrators: To understand catalogue, inventory, and order management functionality.**
- **Customers: To represent their expected interactions and requirements with the system.**
- **Project Evaluators: To evaluate whether the implemented system satisfies the specified requirements.**

**Readers unfamiliar with the project should first refer to Section 2: Overall Description, followed by Section 5: System Features for detailed functionality and Section 6: Nonfunctional Requirements for quality and security expectations.**

### 1.4 Definitions

**Online Bookstore: A web-based system that allows customers to browse, purchase, and track books online.**

**Customer: A registered user who can browse books, manage a shopping cart, place orders, make payments, and track purchases.**

**Administrator: An authorized user responsible for managing books, inventory, catalogue information, and customer orders.**

**Book Catalogue: A collection of books available for purchase, containing details such as title, author, ISBN, category, price, description, and availability.**

**ISBN: International Standard Book Number used to uniquely identify a specific book or edition.**

**Inventory: The collection of records representing the quantity of each book currently available for purchase.**

**Shopping Cart: A temporary collection of books selected by a customer before completing the purchase.**

**Checkout: The process of reviewing cart items, providing delivery details, and proceeding with the payment and order placement.**

**Order: A confirmed request made by a customer to purchase one or more books.**

**Order ID: A unique identifier assigned to an order for identification and tracking purposes.**

**Payment Gateway: A service that securely processes online payment transactions and communicates the payment result to the bookstore.**

**Order Status: The current state of an order, such as Pending, Confirmed, Processing, Shipped, Delivered, or Cancelled.**

**Authentication: The process of verifying the identity of a user before allowing access to protected system functionality.**

**Authorization: The process of determining whether an authenticated user has permission to perform a particular operation.**

**RBAC: Role-Based Access Control, a security mechanism that restricts system functionality according to the user's assigned role.**

**API: Application Programming Interface used to enable communication and data exchange between different software components.**

**CRUD: Create, Read, Update, and Delete operations used to manage system data.**

**Functional Requirement (FR): A requirement that describes a specific function, behaviour, or capability that the system shall provide.**

**Non-Functional Requirement (NFR): A requirement that specifies system qualities or constraints such as performance, security, reliability, usability, or maintainability.**

**RTM: Requirements Traceability Matrix used to map requirements to their corresponding system modules and test cases.**

**Test Case (TC): A defined set of inputs, conditions, and expected results used to verify a system requirement.**

### 1.4 References

- **IEEE Std 830-1998, *IEEE Recommended Practice for Software Requirements Specifications*, used as the reference for preparing and organizing this Software Requirements Specification.**
- **Online Bookstore Project Proposal, used as the primary reference for defining the system objectives, scope, features, user roles, and functional requirements.**
- **Software Engineering Course Materials, PES University, used as a reference for software requirements engineering concepts, requirement specification, UML modelling, and software quality attributes.**

## 2. Overall Description

### 2.1 Product Perspective

**The Online Bookstore is a web-based application that provides customers with a convenient platform to browse, search, purchase, and track books online. The system interacts with customers, administrators, the application database, and payment services. It consists of a user interface, application logic, database management system, and supporting services for authentication, order processing, payment, and inventory management.**

## 2.2 Major Product Functions (Detailed)

- **User registration, login, authentication, and profile management**
- **Browse and search books by title, author, category, or ISBN**
- **Filter and sort books based on price, category, author, and availability**
- **View detailed information about selected books**
- **Add, update, and remove books from the shopping cart**
- **Checkout with delivery address and order summary**
- **Process and verify online payments**
- **Generate and manage unique customer orders**
- **View previous and current orders**
- **Track order status from confirmation to delivery**
- **Cancel orders according to applicable business rules**
- **Administrative management of books and inventory**
- **Automatic stock updates after successful purchases**
- **Administrative viewing and management of customer orders**

## 2.3 User Roles and Characteristics (Expanded)

- **Customer: Has basic knowledge of using web applications and expects a simple, clear, and efficient interface for searching books, purchasing them, making payments, and tracking orders.**
- **Administrator: Responsible for managing books, prices, categories, inventory quantities, and customer orders. Requires access to administrative functions for maintaining the bookstore.**
- **Payment Service: External service responsible for securely processing and verifying customer payment transactions.**

## 2.4 Operating Environment

**The Online Bookstore operates as a web-based application accessible through modern web browsers on desktops, laptops, tablets, and smartphones. The system requires an internet connection and communicates with a backend server and database for storing and retrieving user, book, inventory, payment, and order information.**

## 2.5 Constraints

- **The system requires a stable internet connection for browsing, ordering, and payment processing.**
- **Customer authentication and account information must be securely handled.**
- **Payment transactions must be processed through a secure payment mechanism.**
- **The system must maintain consistency between successful orders and available inventory.**
- **Access to administrative functions must be restricted to authorized administrators.**
- **The application must use a database system capable of securely storing customer, book, inventory, and order information.**

## 3. External Interface Requirements

### 3.1 User Interfaces

**The Online Bookstore shall provide a simple, responsive, and user-friendly web interface through which customers and administrators can interact with the system.**

- **Customer Interface: Provides pages for registration, login, book browsing, searching, filtering, book details, shopping cart, checkout, payment, and order tracking.**
- **Administrator Interface: Provides authorized administrators with interfaces for managing books, prices, categories, inventory, and customer orders.**
- **Navigation: The system shall provide clear menus, buttons, search options, and navigation controls to allow users to move between different functions easily.**
- **Input Validation: The interface shall provide appropriate messages when users enter invalid, incomplete, or incorrect information.**
- **Responsive Design: The interface shall be usable on desktops, laptops, tablets, and smartphones.**
- **Accessibility: Text, controls, labels, and error messages should be presented clearly to support users with different levels of technical familiarity.**

### 3.2 Hardware Interfaces

**The Online Bookstore is primarily a software-based web application and does not require specialized hardware for normal operation.**

- **Client Devices: The system shall support access through desktops, laptops, tablets, and smartphones.**
- **Display: The user interface shall adapt to different screen sizes and resolutions.**
- **Input Devices: Customers may interact with the system using a keyboard, mouse, touchscreen, or other standard input devices.**
- **Server Hardware: The application and database shall operate on server hardware capable of supporting concurrent users, database operations, and order processing.**
- **Storage: Server-side storage shall be used for maintaining application data such as customer accounts, book information, inventory, and order records.**

### 3.3 Software Interfaces

**The Online Bookstore shall communicate with several software components to provide its major functionality.**

- **Database Management System: Used to store and retrieve customer accounts, book details, inventory information, shopping carts, payments, and order records.**
- **Payment Gateway/API: Used for securely processing online payment requests and receiving payment success or failure responses.**
- **Authentication Module: Used to validate customer and administrator credentials and control access to protected functions.**
- **Application Server: Handles business logic, user requests, order processing, inventory updates, and communication between the user interface and database.**
- **Web Browser: The system shall provide compatibility with commonly used modern web browsers.**
- **Logging and Monitoring Services: Where implemented, application logs shall record relevant system events, errors, and administrative activities for monitoring and troubleshooting.**

**3.4 Communications Interfaces**

- **The system shall use HTTPS with TLS for secure communication between the client browser and the application server.**
- **Communication between the application server and database shall use secure database connections where supported.**
- **Communication with external payment services shall use secure API protocols such as HTTPS.**
- **The system shall validate responses received from external services before updating payment or order status.**
- **The system shall handle communication failures gracefully and display appropriate messages to the customer.**
- **Failed requests shall not result in duplicate orders or inconsistent payment and inventory records.**
- **The system shall maintain transaction consistency when communicating between order processing, payment, and inventory components.**

**Security Objectives**

**The Online Bookstore shall satisfy the following major security objectives:**

**SO-1: Confidentiality**

**The system shall protect sensitive customer information, account credentials, delivery information, and payment-related information from unauthorized access or disclosure.**

**SO-2: Integrity and Accountability**

**The system shall ensure that customer, order, payment, and inventory information cannot be modified by unauthorized users and that important system activities can be traced to the responsible user or administrative action.**

**Security Requirements**

**SEC-1:SecureAuthentication**

**The system shall require customers and administrators to authenticate using valid credentials before accessing protected functions. Administrative functions shall only be available to authorized administrators.**

**SEC-2:PasswordProtection**
**The system shall not store customer passwords in plain text. Passwords shall be protected using an appropriate secure password-hashing mechanism.**

**SEC-3:AuthorizationandAccessControl**
**The system shall enforce role-based access control so that customers can access only customer-related functions, while administrative operations are restricted to authorized administrators.**

**SEC-4:SecureCommunication**
**The system shall use HTTPS/TLS for communication involving login credentials, customer information, checkout, payment requests, and other sensitive data.**

**SEC-5:Data and Transaction Integrity**
**The system shall validate user inputs and securely process order, payment, and inventory operations to prevent unauthorized modification, duplicate transactions, or inconsistent records.**

**SEC-6: Session Security**
**The system shall securely manage authenticated user sessions and invalidate the session when the user logs out.**

**SEC-7: Payment Data Protection**
**The system shall avoid unnecessarily storing sensitive payment credentials and shall rely on the payment service's secure mechanisms for processing payment information.**

**SEC-8: Administrative Activity Protection**
**The system shall restrict modification of book details, prices, inventory, and customer orders to authenticated and authorized administrators.**

## 4. Analysis Models

### 4.1 Use Case Model

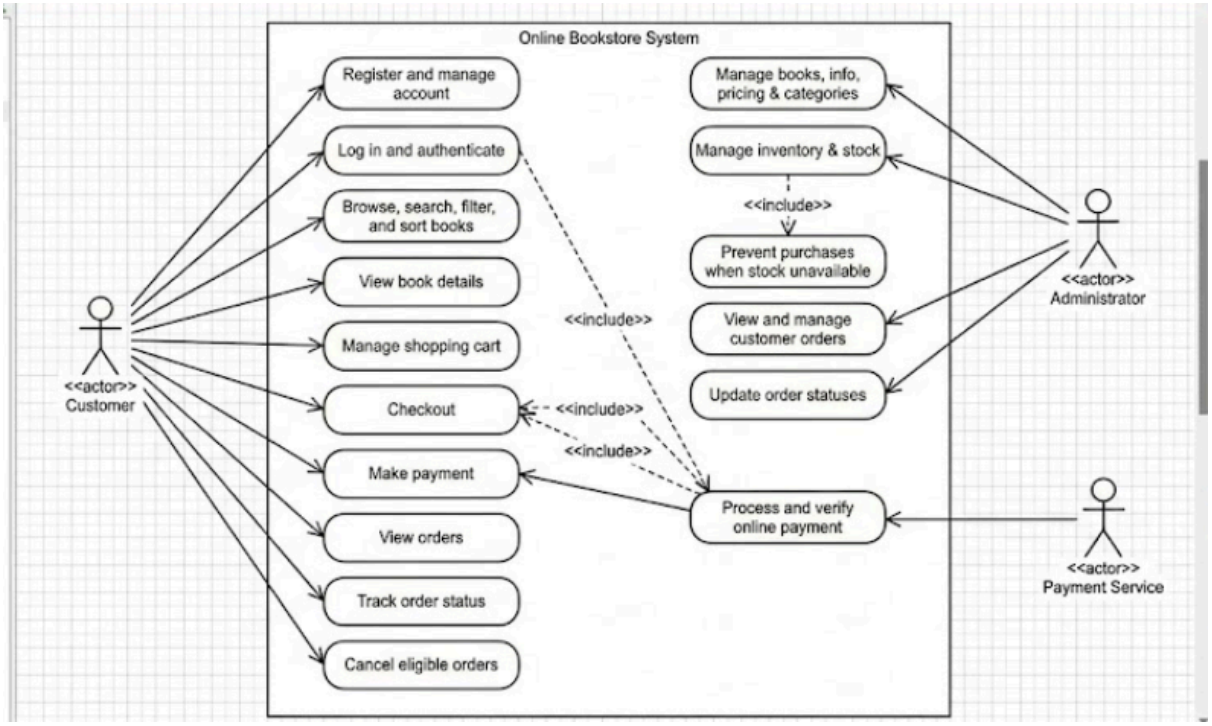
**The Online Bookstore has three primary actors: Customer, Administrator, and Payment Service. The use-case model describes how these actors**

**interact with the system to complete the major functions identified in the requirements.**

**Customer: Register and manage account; log in and authenticate; browse, search, filter, and sort books; view book details; manage the shopping cart; checkout and provide delivery information; make payments; view orders; track order status; and cancel eligible orders.**

**Administrator: Manage books, book information, pricing and categories; manage inventory and stock quantities; view and manage customer orders; update order statuses; and prevent purchases when sufficient stock is unavailable.**

**Payment Service: Process and verify online payment transactions and return payment success or failure information to the bookstore.**




## 4.2 Domain Model

**The domain model is derived from the entities and responsibilities described in the SRS. The main domain entities are Customer, Book, Book Catalogue, Inventory, Shopping Cart, Order, Payment, Administrator, and Payment Gateway.**

**Customer owns account and delivery information, maintains a shopping cart, and places orders.**

**Book represents a purchasable title and contains information such as title, author, ISBN, category, price, description, and availability.**

**Book Catalogue groups the available books and supports browsing, searching, filtering, and sorting.**

**Inventory maintains the quantity of each book available for purchase and is updated after successful purchases.**

**Shopping Cart contains books selected by a customer before checkout.**

**Order records a confirmed purchase, has a unique Order ID, contains one or more books, and has an Order Status.**

**Payment records the payment result associated with checkout, while the Payment Gateway processes the transaction.**

**Administrator manages catalogue, inventory, and customer orders and is restricted to authorized administrative functions.**

### 4.3 Activity Model

**The primary customer activity flow is: authenticate or register → browse/search catalogue → select a book → add or update the shopping cart → checkout and provide delivery details → process and verify payment → create the order and Order ID → update inventory → display order confirmation → allow order tracking and, where permitted, cancellation.**

**The administrative activity flow is: authenticate as administrator → manage books and catalogue information or review customer orders → update inventory or order status → ensure stock and transaction consistency → save the updated records.**

### 4.4 Sequence Model

**For a purchase, the Customer interacts with the web interface, which sends requests to the Application Server. The Application Server validates the request, accesses the Database Management System, communicates with the Payment Gateway/API, and then updates the order and inventory records. A successful payment results in order creation, inventory update, and confirmation to the Customer. Failed communication or payment must not create duplicate orders or inconsistent payment and inventory records.**

## 5. System Features

### 5.1 Customer Account Management

**The system shall provide customer registration, secure login, authentication, authorization, and profile management. Customers shall be able to maintain their account information and access protected bookstore functions only after successful authentication.**

**5.2 Book Catalogue and Search**

**The system shall allow customers to browse the book catalogue, search by title, author, category, or ISBN, filter and sort by relevant attributes such as price, category, author, and availability, and view detailed information about selected books.**

**5.3 Shopping Cart**

**The system shall allow customers to add books to the shopping cart, update item quantities, and remove items. The cart shall represent the temporary collection of books selected before checkout.**

**5.4 Checkout and Delivery**

**The system shall provide a checkout process in which customers review cart items, provide delivery information, and review an order summary before proceeding with payment.**

**5.5 Payment Processing**

**The system shall process and verify online payments through the payment gateway/API. Payment responses shall be validated before payment or order status is updated, and failed requests shall not result in duplicate orders or inconsistent records.**

**5.6 Order Management**

**The system shall generate a unique Order ID for confirmed purchases, allow customers to view previous and current orders, track order status from confirmation to delivery, and cancel orders when permitted by applicable business rules.**

**5.7 Inventory Management**

**The system shall maintain stock quantities for books and automatically update inventory after successful purchases. The system shall prevent purchases when sufficient stock is unavailable and shall maintain consistency between orders and available inventory.**

**5.8 Administrator Management**

**The system shall provide authorized administrators with facilities to add, update, and remove books; manage book information, pricing, and categories; manage inventory; view and manage customer orders; and update order statuses.**

**5.9 Order Tracking and Status**

**The system shall provide the current state of each order using defined order statuses such as Pending, Confirmed, Processing, Shipped, Delivered, or Cancelled, and shall make relevant status information available to the customer.**

**5.10 Security and Access Control**

**The system shall enforce authentication and role-based authorization for customer and administrator functions. Sensitive information and payment-related operations shall be protected using secure communication and appropriate security controls.**

**Security Objectives**

**The Online Bookstore shall satisfy the following major security objectives:**

**SO-1: Confidentiality**
**The system shall protect sensitive customer information, account credentials, delivery information, and payment-related information from unauthorized access or disclosure.**

**SO-2: Integrity and Accountability**
**The system shall ensure that customer, order, payment, and inventory information cannot be modified by unauthorized users and that important system activities can be traced to the responsible user or administrative action.**

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**The system shall enforce role-based access control so that customers can access only customer-related functions, while administrative operations are restricted to authorized administrators.**

**SEC-4:SecureCommunication**
**The system shall use HTTPS/TLS for communication involving login credentials, customer information, checkout, payment requests, and other sensitive data.**

**SEC-5:Data and Transaction Integrity**
**The system shall validate user inputs and securely process order, payment, and inventory operations to prevent unauthorized modification, duplicate transactions, or inconsistent records.**

**SEC-6: Session Security**
**The system shall securely manage authenticated user sessions and invalidate the session when the user logs out.**

**SEC-7: Payment Data Protection**
**The system shall avoid unnecessarily storing sensitive payment credentials and shall rely on the payment service's secure mechanisms for processing payment information.**

**SEC-8: Administrative Activity Protection**
**The system shall restrict modification of book details, prices, inventory, and customer orders to authenticated and authorized administrators.**

## 6. Non-Functional Requirements

### 6.1 Performance Requirements

**PERF-1: Response Time — The system shall return search and catalogue browsing results within 2 seconds under normal load conditions.**
**PERF-2: Checkout Performance — The system shall complete cart-to-order transitions (excluding external payment gateway processing time) within 3 seconds.**
**PERF-3: Concurrent Users — The system shall support a minimum of 500 concurrent users without significant degradation in response time.**
**PERF-4: Page Load Time — Book catalogue and product detail pages shall load within 3 seconds on a standard broadband connection.**

### 6.2 Scalability Requirements

**SCAL-1: Catalogue Growth — The system shall support scaling of the book catalogue to at least 100,000 titles without requiring architectural**

**redesign.**

**SCAL-2: Load Scaling — The system architecture shall support horizontal scaling of application servers to accommodate increased traffic during peak periods (e.g., sales, festive seasons).**

**SCAL-3: Database Scaling — The database shall be designed to accommodate growth in customer accounts, order history, and inventory records over time.**

### 6.3 Reliability and Availability Requirements

**REL-1: System Uptime — The system shall maintain an uptime of at least 99% excluding scheduled maintenance windows.**

**REL-2: Fault Tolerance — In the event of a payment gateway or third-party service failure, the system shall roll back the transaction and notify the customer without corrupting order or inventory records.**

**REL-3: Data Backup — The system shall perform regular automated backups of customer, order, inventory, and catalogue data.**

**REL-4: Recovery — The system shall be capable of restoring service and data to the most recent consistent backup point in the event of a failure.**

### 6.4 Usability Requirements

**USE-1: Ease of Navigation — A first-time customer shall be able to locate and view a book's details within 3 clicks from the homepage.**

**USE-2: Error Messaging — The system shall present clear, non-technical error and validation messages to guide users toward correct input.**

**USE-3: Consistency — The system shall maintain consistent layout, terminology, and interaction patterns across all customer-facing and administrative pages.**

**USE-4: Learnability — A new administrator shall be able to perform basic catalogue and inventory management tasks without requiring external training, using in-app guidance/labels alone.**

### 6.5 Maintainability and Supportability Requirements

**MAIN-1: Modularity — The system shall be designed with a modular architecture separating catalogue, cart, checkout, payment, order, and administrative modules to simplify maintenance and future updates.**

**MAIN-2: Logging — The system shall log significant errors and administrative actions to support debugging and maintenance without exposing sensitive customer data in logs.**

**MAIN-3: Code Documentation — Core modules shall be documented sufficiently to allow a new developer to understand and modify functionality without requiring the original author.**

### 6.6 Portability Requirements

**PORT-1: Browser Compatibility — The system shall function correctly on the latest stable versions of Chrome, Firefox, Safari, and Edge.**

**PORT-2: Device Compatibility — The system shall render correctly across desktop, tablet, and mobile screen sizes without loss of core functionality.**

## 7. Other Requirements

### 7.1 Legal and Regulatory Requirements

**LEG-1: Data Protection — The system shall handle customer personal and payment-related data in a manner consistent with applicable data protection principles (e.g., data minimization, secure storage).**

**LEG-2: Payment Compliance — Payment processing shall be handled through a PCI-DSS-compliant third-party payment gateway; the system itself shall not store raw card details.**

**LEG-3: Terms of Use — The system shall require customers to accept the platform's terms of service and privacy policy during registration.**

### 7.2 Database Requirements

**DB-1: Data Retention — Order and transaction records shall be retained for a minimum period sufficient to support customer service, returns, and auditing needs.**

**DB-2: Referential Integrity — The database shall enforce referential integrity between customers, orders, order items, books, and inventory records.**

**DB-3: Consistency — The database shall ensure that inventory counts remain consistent with confirmed orders at all times, even under concurrent access.**

### 7.3 Business Rules

**BR-1: Stock Enforcement — An order shall not be confirmed if the requested quantity of any book exceeds available stock at the time of checkout.**

**BR-2: Order Cancellation Window — Customers shall only be permitted to cancel an order before it reaches "Shipped" status.**

**BR-3: Pricing Authority — Only administrators shall have the authority to modify book prices; price changes shall not retroactively affect already-confirmed orders.**

### 7.4 Assumptions and Dependencies

- **The system assumes availability of a reliable third-party payment gateway API for transaction processing.**
- **The system assumes customers have access to a modern web browser and stable internet connectivity.**
- **The system depends on the accuracy of inventory data provided/updated by administrators.**
- **Delivery/courier tracking, where referenced, is assumed to be handled by an external logistics provider and is out of scope for internal implementation.**